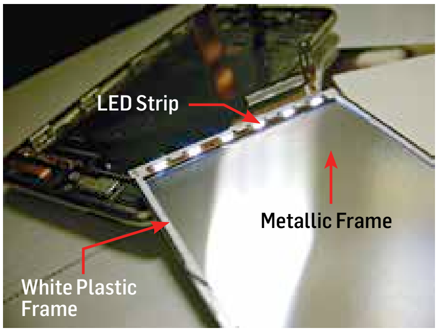
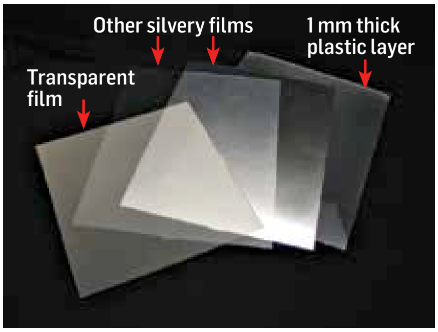
Glucose monitoring

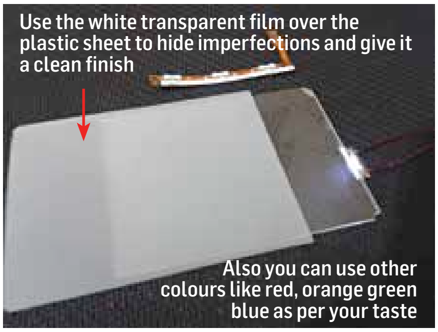
Two engineering dads have created Project Nightscout to continuously monitors glucose levels and sends the data to the cloud **http://dgit.in/RealTimeGlu**

Portable Pi Laptop

Here's a quick and easy DIY on building a portable Raspberry Pi laptop from Instructables: **http://dgit.in/RaspPiLap**

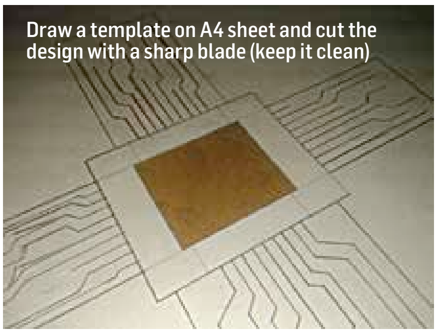




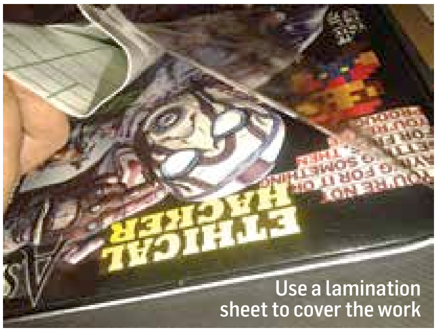


Use the white transparent film over the plastic sheet to hide imperfections and give it a clean finish

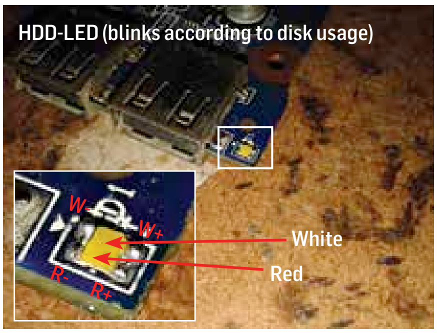
Also you can use other colours like red, orange green blue as per your taste

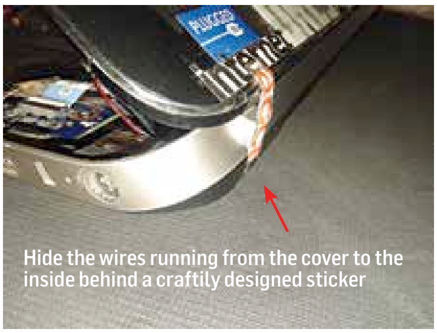


Draw a template on A4 sheet and cut the design with a sharp blade (keep it clean)



Use a lamination sheet to cover the work





Hide the wires running from the cover to the inside behind a craftily designed sticker

layer it onto the top of the thick plastic sheet. On the bottom, affix the silvery sheet which will reflect more light through the upper sheet. The size of this contraption should be a guiding factor for your cover design.

2 Prepare the cover

Grab a few A4 sized paper sheets and draw a pattern which you'll be cutting out. The dimensions of the section that is to be cut should be lesser than the light contraption we prepared in the previous section. Now you've reached the creative part where you design the cover using various magazine cutouts and what not. Once you've finalised the design you should go ahead and layer it over the illumination contraption. Check all the connections and glue it all together while maintaining an even thickness across the entire design. Once the glue dries off, laminate the entire setup (design+illuminated panel).

3 Hook up the wires

Open up your laptop and figure out where the status LEDs are. Check the voltage levels of the LEDs since you can't draw more than what it's currently drawing. Drawing more current will cause your entire setup to run dimmer than required. In such a scenario you'll need to use a transistor (BC478) as a switch (Forward-biased configuration) and source the current from a 5V source. Refer this site for instructions (http://dgit.in/DIYLapCov2). Hot wire your illumination panel's terminals to the status LEDs on the motherboard's laptop. Be careful about the polarity of the terminals.

4 Hide your wires

Laptops are built differently and the same approach might not work out for all. The last step is to route the wires in an inconspicuous manner. Take a look at how Sandeep has wired his cover in the last image and check if you can work out a similar solution.

5 That's it!

Let us know how this works out for you, we'd love to see more similar creative DIYs. Send in your creations to editor@digit.in. Meanwhile you can check out Sandeep's work over at http://dgit.in/DIYLapCov3.